Lab 1 - Setting Up a New Drawing

Course: WSQ Technical Drawing with AutoCAD (TGS-2021005538) | Topic 01: Getting Started with AutoCAD

Objective: LO1 - read and visualize AutoCAD technical drawings; set up a drawing environment (A1, K1)

Goal

Create and verify a metric drawing environment, compare saved workspace and drafting-setting examples, then open a finished drawing and practise reading it with zoom, pan and ViewCube.

You'll produce

A correctly configured metric drawing saved as DWG, a settings audit, and confident navigation of an existing technical drawing.

Commands / tools: ACADISO template, UNITS, LIMITS, DSETTINGS, WSCURRENT, ZOOM, PAN, ViewCube

Files in this lab

- lab01-table.dwg - AutoCAD drawing file
- reading-practice.dwg - AutoCAD drawing file
- workspaces.dwg - AutoCAD drawing file
- drafting-settings.dwg - AutoCAD drawing file

Step-by-step

1. Start AutoCAD and create a new drawing from the ISO metric template

```
NEW → acadiso.dwt
```

2. Open workspaces.dwg, compare the available workspaces, then switch to Drafting & Annotation

```
OPEN workspaces.dwg → WSCURRENT → Drafting & Annotation
```

3. Set decimal millimetres with 0.00 precision and decimal degrees for angles

```
UNITS → Insertion scale: Millimetres → Length: Decimal, 0.00
```

4. Frame an A3-sized working area with drawing limits, then zoom to it

```
LIMITS → 0,0 → 420,297  |  ZOOM → All
```

5. Open drafting-settings.dwg and audit grid, snap, ortho, polar and object-snap settings

```
OPEN drafting-settings.dwg → DSETTINGS | F7 | F9 | F8 | F10 | F3
```

6. Open lab01-table.dwg and navigate it: wheel-zoom, pan, zoom window and zoom extents

```
OPEN lab01-table.dwg → Z → W / Z → E
```

7. Identify the views, dimensions and title information in the drawing, then save your setup file

```
SAVEAS → lab01-setup.dwg
```

Check your work

INSUNITS reports millimetres, limits are 420 × 297, Drafting & Annotation is current, the precision-aid audit is complete, and lab01-setup.dwg reopens with the settings intact.